

IMDS SAAS PLATFORM FRONTEND ARCHITECTURE & UI/UX SPECIFICATION

Project Structure, Portal Designs, Page Inventory & Feature Map

Based on: Unified Architecture Document v2.1 | Nissan RFI v1.9 | All 10 Modules + 6 AI Innovations

Prepared for: Nissan Automotive Europe
Prepared by: Marklytics Ltd
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1. Frontend Project Structure

The frontend lives in `apps/web/` within the Turborepo monorepo. It is a Next.js 15+ application using the App Router with React Server Components, Tailwind CSS 4, and a shared UI component library from `packages/ui/`.

1.1 Complete Frontend Directory Tree

```

apps/web/ | public/ | icons/ | SVG icons, favicons, | images/
# Static images, logos | locales/ | # i18n translation JSON (26
languages) | src/ | app/ | # Next.js App Router | (auth)/
# Auth group (login, register, forgot-password) | login/
register/ | forgot-password/ | sso-callback/
layout.tsx | (portal)/ | # Authenticated portal group
supplier/ | # Supplier Portal | dashboard/ | mds/
| new/ | [id]/ | page.tsx
| sds/ | upload/ | [id]/
| page.tsx | sui/ | documents/
requests/ | profile/ | layout.tsx | oem/
# OEM Portal (Nissan) | dashboard/ | mds/
| review/ | [id]/ | page.tsx
| bom/ | analysis/ | upload/
| [id]/ | sds/ | scip/ | reach/
| clp/ | rrr/ | sustainability/
| pcf/ | dpp/ | cbam/
| csrd/ | suppliers/ | chasing/
| [id]/ | page.tsx | reports/
integrations/ | ai-insights/ | layout.tsx
admin/ | # Platform Admin Portal | dashboard/
tenants/ | users/ | billing/ | system/
| ai-models/ | audit-logs/ | layout.tsx
| compliance/ | # Compliance Officer Portal | dashboard/
| regulatory/ | forecasting/ | dpp/
| audit/ | layout.tsx | layout.tsx | # Shared
authenticated layout | api/ | # Next.js API routes (BFF)
| auth/ | mds/ | bom/ | sds/
webhooks/ | layout.tsx | # Root layout (providers, fonts)
globals.css | not-found.tsx | components/ | # App-level components
| layouts/ | AppShell.tsx | # Main layout (sidebar + header +
content) | Sidebar.tsx | TopBar.tsx
Breadcrumbs.tsx | CommandPalette.tsx | mds/
MdsTreeBuilder.tsx | SubstanceEntry.tsx | MdsValidation.tsx
| MdsVersionDiff.tsx | MdsStatusBadge.tsx | bom/
BomUploader.tsx | BomAnalysisTable.tsx | BomDrillDown.tsx
| ComplianceHeatmap.tsx | sds/ | SdsUploader.tsx
SdsViewer.tsx | SdsWorkflow.tsx | SdsTranslation.tsx
scip/ | ScipDossierBuilder.tsx | TarcCodeManager.tsx
| ScipSubmissionStatus.tsx | reach/ | RegistrationTracker.tsx
| VolumeMonitor.tsx | RestrictionList.tsx
SvhcCalculator.tsx | clp/ | HazardClassifier.tsx
LabelGenerator.tsx | PcndossierBuilder.tsx | ClpSearchDisplay.tsx
| rrr/ | RrrCalculator.tsx | MaterialClassification.tsx
| RecycledContentTracker.tsx | sustainability/
PcfDashboard.tsx | DppGenerator.tsx | CbamTracker.tsx
| CsrdExport.tsx | ai/ | CompliancePrediction.tsx
MaterialClassifierAI.tsx | RegulatoryAlerts.tsx
PredictiveAnalytics.tsx | QueryAssistant.tsx
AiExplainability.tsx | suppliers/ | SupplierList.tsx
ChasingEngine.tsx | ResponseFunnel.tsx | SupplierProfile.tsx
| dashboards/ | KpiCard.tsx | DeadlineTimeline.tsx
RiskTrajectory.tsx | ComplianceFunnel.tsx | reports/
ReportBuilder.tsx | ExportEngine.tsx | ScheduledReports.tsx
| common/ | DataTable.tsx | FileUpload.tsx
StatusBadge.tsx | EmptyState.tsx | ConfirmDialog.tsx
| NotificationBell.tsx | hooks/ | useAuth.ts | useMds.ts
| useBom.ts | useSds.ts | useSupplier.ts | useCompliance.ts
| useAiPrediction.ts | useRealtime.ts | # WebSocket for live updates
| useDebounce.ts | useInfiniteScroll.ts | lib/ | api-client.ts
# Auto-generated typed client | auth.ts | # Cognito/JWT helper
| i18n.ts | # 26-language setup | permissions.ts | # RBAC/ABAC
client-side guards | websocket.ts | # WS connection manager
analytics.ts | # PostHog/Mixpanel | utils.ts | stores/
# Zustand state management | authStore.ts | mdsStore.ts
bomStore.ts | notificationStore.ts | uiStore.ts | styles/

```

```

├─ globals.css | ┌─ theme.ts # Tailwind theme tokens | ┌─ types/ |
├─ mds.ts | ┌─ bom.ts | ┌─ sds.ts | ┌─ supplier.ts | ┌─
├─ compliance.ts | ┌─ ai.ts | ┌─ api.ts | ┌─ next.config.ts | ┌─ tailwind.config.ts
├─ tsconfig.json | ┌─ .env.local | ┌─ package.json

```

1.2 Technology Stack (Frontend)

Layer	Technology	Version	Purpose
Framework	Next.js (App Router)	15.x	RSC, SSR, file-based routing, API routes as BFF
Language	TypeScript	5.x	Type safety across entire frontend
Styling	Tailwind CSS	4.x	Utility-first responsive styling with design tokens
Components	shadcn/ui + Radix Primitives	Latest	Accessible, unstyled headless components
State (Client)	Zustand	5.x	Lightweight global state for UI/session
Server State	TanStack Query (React Query)	5.x	Caching, refetching, optimistic updates
Forms	React Hook Form + Zod	7.x / 3.x	Performant forms with schema validation
Tables	TanStack Table	8.x	Virtualised, sortable, filterable data tables
Charts	Recharts + D3.js	2.x / 7.x	Compliance heatmaps, PCF charts, trend lines
Drag & Drop	dnd-kit	6.x	MDS tree builder drag-and-drop
i18n	next-intl	3.x	26-language support (RFI §17.3)
Real-time	Socket.IO client	4.x	Live notifications, BOM analysis progress
Testing	Vitest + Playwright + Testing Library	Latest	Unit, integration, E2E testing
API Client	Auto-generated (openapi-typescript)	Latest	Type-safe API calls from OpenAPI spec
Auth	next-auth + AWS Cognito	5.x	SAML SSO, JWT, session management
Analytics	PostHog	Latest	Feature flags, session replay, funnel tracking
Icons	Lucide React	Latest	Consistent icon library (500+ icons)

1.3 Shared UI Component Library (packages/ui/)

All portals share a common design system built on shadcn/ui + Radix, extended with domain-specific components:

```

packages/ui/ ┌─ src/ | ┌─ primitives/ # Base: Button, Input, Select, Dialog,
etc. | ┌─ composite/ # DataTable, FileUpload, DateRangePicker | ┌─ domain/
# MdsTree, SubstanceChip, ComplianceBadge | ┌─ charts/ # HeatmapChart,
TimelineChart, FunnelChart | ┌─ layouts/ # AppShell, PageHeader, EmptyState |
┌─ tokens/ # colors.ts, spacing.ts, typography.ts | ┌─ stories/
# Storybook stories per component ┌─ package.json

```

2. Design System & Visual Language

2.1 Colour Palette

Token	Hex	Usage
primary-900	#1B2A4A	Headings, sidebar background, primary text
primary-600	#2563EB	Links, active states, primary buttons
primary-100	#DBEAFE	Selected row backgrounds, focus rings
teal-600	#0D9488	Accent, secondary buttons, success-adjacent
teal-100	#CCFBF1	Teal highlights, card backgrounds
success-600	#16A34A	Compliant status, accepted MDS, pass
success-100	#D1FAE5	Success backgrounds
warning-600	#D97706	Pending review, approaching deadlines
warning-100	#FEF3C7	Warning backgrounds
danger-600	#DC2626	Non-compliant, overdue, SVHC alerts, rejected
danger-100	#FEE2E2	Danger backgrounds
neutral-50	#F9FAFB	Page background
neutral-200	#E5E7EB	Borders, dividers
neutral-500	#6B7280	Secondary text, placeholders
neutral-900	#111827	Primary body text

2.2 Typography Scale

Element	Font	Size	Weight	Line Height
Page Title	Inter	30px / 1.875rem	700 Bold	1.2
Section Title	Inter	24px / 1.5rem	600 Semi	1.3
Card Title	Inter	18px / 1.125rem	600 Semi	1.4
Body	Inter	14px / 0.875rem	400 Regular	1.6
Small / Caption	Inter	12px / 0.75rem	400 Regular	1.5
Monospace (data)	JetBrains Mono	13px / 0.8125rem	400 Regular	1.5
Table Header	Inter	12px / 0.75rem	600 Semi / Uppercase	1.5

2.3 Spacing System (8px base grid)

All spacing uses multiples of 8px. Key tokens: xs=4px, sm=8px, md=16px, lg=24px, xl=32px, 2xl=48px, 3xl=64px. Component padding: Buttons=8px 16px, Cards=16px 20px, Page sections=32px 0px.

2.4 RAG Compliance Status System

The platform uses a consistent Red-Amber-Green (RAG) system across all portals:

Status	Colour	Badge Text	Meaning
Compliant	Green (#16A34A)	COMPLIANT	All rules pass; MDS accepted; substance below threshold
Review Needed	Amber (#D97706)	REVIEW	Pending human review; AI flagged uncertainty; approaching deadline

Non-Compliant	Red (#DC2626)	ALERT	Rule violation; SVHC above limit; overdue; rejected MDS
In Progress	Blue (#2563EB)	PENDING	Awaiting data; submission in draft; analysis running
AI Predicted	Purple (#7C3AED)	AI	AI prediction, not yet confirmed by human review

Leaf	Sustainability	/oem/sustainability	PCF dashboard, DPP, CBAM, CSRD, Catena-X
Users	Suppliers	/oem/suppliers	List, chasing engine, response funnel, [id]/profile
BarChart3	Reports	/oem/reports	Standard reports, builder, scheduled, export
Link	Integrations	/oem/integrations	IMDS, SAP, Oracle, ECHA, InfoDyne, Nissan BOM
Brain	AI Insights	/oem/ai-insights	Predictions, alerts, analytics, query assistant
Settings	Settings	/oem/settings	Org config, users, roles, API keys, webhooks

C) Admin Portal Sidebar

Icon	Label	Route	Sub-pages
LayoutDashboard	Dashboard	/admin/dashboard	Platform health, tenant stats, revenue
Building	Tenants	/admin/tenants	List, [id]/config, provisioning, DB isolation
Users	Users	/admin/users	All users, roles, permissions, SAML config
CreditCard	Billing	/admin/billing	Subscriptions, invoices, usage metering
Server	System	/admin/system	Health, queues, jobs, feature flags
Brain	AI Models	/admin/ai-models	Model registry, versions, A/B tests, governance
ScrollText	Audit Logs	/admin/audit-logs	Immutable event log, search, export

D) Compliance Officer Portal Sidebar

Icon	Label	Route	Sub-pages
LayoutDashboard	Dashboard	/compliance/dashboard	Executive view, risk scores, coverage
AlertTriangle	Regulatory	/compliance/regulatory	Alerts, change tracker, impact analysis
TrendingUp	Forecasting	/compliance/forecasting	AI predictions, risk trajectory, deadlines
Fingerprint	DPP	/compliance/dpp	Digital Product Passport management, issuance
ClipboardCheck	Audit	/compliance/audit	Compliance audit trail, evidence, certificates

4. Supplier Portal — Page-by-Page Specification

7 pages + sub-pages. Target users: 4,000+ supplier companies submitting compliance data to Nissan.

4.1 Supplier Dashboard

Dashboard Route: `/supplier/dashboard`

Landing page after login. Shows supplier's compliance status at a glance.

Components:

- KPI Row (4 cards): Open Requests (count), MDS Submitted (this month), SDS Pending (count), Compliance Score (%).
- Action Items Panel: Outstanding OEM requests sorted by deadline urgency. Click to navigate directly to task.
- Recent Activity Feed: Timeline of recent submissions, approvals, rejections, chasing messages.
- Deadline Calendar: Mini calendar highlighting upcoming submission deadlines. Colour-coded by urgency.
- Notification Centre: Unread messages from OEM, system alerts, regulatory updates.

4.2 MDS Management

MDS List Route: `/supplier/mds`

Table of all MDS created/submitted by this supplier. Filterable by status, part number, date.

Features:

- DataTable: Columns = Part Number, MDS ID, Status (Draft/Submitted/Accepted/Rejected), Version, Last Updated, Compliance Score. Sortable, filterable, searchable.
- Quick Actions: New MDS, Import from IMDS, Duplicate, Export XML.
- Status Filters: Tabs for All / Draft / Submitted / Accepted / Rejected / Needs Update.
- Bulk Actions: Select multiple → Submit All / Export / Delete Draft.

MDS Creator/Editor Route: `/supplier/mds/new` | `/supplier/mds/[id]`

Multi-step wizard for creating or editing a Material Data Sheet. Core workflow of the platform.

Step 1 — Part Selection:

- Part number input (manual or auto-complete from Nissan BOM feed).
- BOM upload: CSV/XML import to populate part tree automatically.
- Indirect parts toggle: Include production/packaging materials (RFI §1.1).
- EEA Import flag: Toggle for products imported from outside EEA (RFI §1.2).

Step 2 — Material Tree Builder:

- Drag-and-drop tree editor: Vehicle → Component → Sub-component → Material → Substance.
- Visual tree with expand/collapse nodes. Each node shows name, weight, and mini-compliance badge.
- Add/remove/reorder nodes. Clone subtrees. Import from existing MDS.
- Service parts assembly: Combine existing MDS into service part trees (RFI §4).
- Nissan-built parts: Assemblies at Nissan sites including aftersales (RFI §5).

Step 3 — Substance Data Entry:

- Per-material substance table: CAS number (auto-lookup), substance name, concentration (min/max %), classification.
- AI Material Classification (AI-2): Upload SDS PDF → auto-extract substances with confidence scores. Manual correction for low-confidence fields.
- SVHC/PFAS Auto-Flag: Real-time highlighting of substances on SVHC candidate list, GADSL, or PFAS restriction list.
- PCF fields (IMDS 15.x): pcf_product (kg CO₂e/kg), pcf_transport (kg CO₂e/delivery), manufacturing site.
- Recycled content fields: Pre/post-consumer %, ELV-origin %.

Step 4 — Document Attachment:

- File upload dropzone: SDS PDF, SUI documents, certificates, CLP labels, letter of compliance.
- SDS → MDS XML linkage: Auto-generate XML from attached SDS containing substance data (RFI §2, §9).
- Workers instructions attachment: Various formats linked to substance/mixture use (RFI §2).
- Document validation: File type checks, virus scan, OCR extraction preview.

Step 5 — Validation & AI Prediction:

- Real-time rule validation: Run against GADSL, ELV, REACH, PFAS rules. Results in side panel with pass/fail per rule.
- AI Compliance Prediction (AI-1): Risk score 0-100 with SHAP explanation showing contributing factors.
- Error panel: Blocking errors (must fix), warnings (should review), suggestions (optional improvements).
- Historical comparison: Diff view against previous MDS version.

Step 6 — Review & Submit:

- Full MDS summary view: All data on one scrollable page for final review.
- Submit button: Locks MDS, sends to OEM review queue, triggers notification.
- Save as Draft: Preserve progress without submitting.

4.3 SDS Management

SDS List & Upload *Route: /supplier/sds*

Manage Safety Data Sheets. Upload PDFs, track approval status.

- DataTable: SDS Name, Substance/Mixture, Status (Draft/Under Review/Approved/Rejected), Language, Upload Date.
- Upload: Drag-drop PDF with OCR extraction preview. AI extracts Section 2 (hazards) and Section 3 (composition).
- Workflow: Draft → Submitted → Under Review → Approved/Rejected (with comments).
- eSDS toggle: Mark as extended SDS with exposure scenarios (REACH >10t/a) (RFI §2).
- Translation: Request AI translation to any of 26 languages. Show translation confidence score.

4.4 SUI Management

Safe Use Instructions *Route: /supplier/sui*

Upload and manage SVHCs with Safe Use Instructions per RFI §3.

- SVHC list with linked SUI documents. Upload interface for new SUI files.
- Link SUI to specific MDS and substance records.

4.5 Documents

Document Library *Route: /supplier/documents*

Central repository for all uploaded documents.

- Grid/list view toggle. Filter by type (SDS, SUI, Certificate, Label, Letter of Compliance).
- Full-text search across document content (OCR-indexed).
- Version history per document. Download, preview, share.

4.6 Requests Inbox

OEM Requests *Route: /supplier/requests*

Inbox of requests from OEM (Nissan) for data submission.

- Priority-sorted list: Overdue (red), Due Soon (amber), On Track (green).
- Each request: Part number, request type (MDS/SDS/SUI), deadline, requester.
- Quick actions: Start MDS, Upload Document, Request Extension.
- Batch response: Reply to multiple requests for same supplier (RFI §17.1).

4.7 Supplier Profile

Profile & Settings *Route: /supplier/profile*

Company profile, contacts, preferences.

- Company details: Name, address, DUNS number, contact persons.
- Notification preferences: Email digest frequency, in-app alert types.
- Language preference: Default UI language from 26 supported.
- API access: View/regenerate API keys for automated submissions.

5. OEM Portal (Nissan) — Page-by-Page Specification

14 top-level sections with 40+ sub-pages. The primary portal for Nissan compliance engineers to manage material compliance across all RFI modules.

5.1 OEM Dashboard

Main Dashboard *Route: /oem/dashboard*

Executive overview of entire compliance program.

Row 1 — KPI Cards (6):

- MDS Coverage: X% of parts with accepted MDS (target: 100%).
- Overdue Submissions: Count of supplier submissions past deadline.
- SVHC Alerts: Active SVHC threshold violations.
- SCIP Pending: SCIP dossiers not yet submitted to ECHA.
- PCF Completeness: % of MDS with PCF data populated.
- PFAS Exposure: Count of parts containing PFAS substances.

Row 2 — Compliance Heatmap:

- Interactive grid: Product lines (rows) × Regulation (columns: GADSL, ELV, REACH, PFAS, SCIP, CLP).
- Cell colour: RAG status. Click to drill into specific product line + regulation.
- PFAS highlighting: Dedicated visual indicator per product line.

Row 3 — Split View:

- Left: Deadline Timeline (Gantt-style): Regulatory deadlines, submission due dates, SCIP windows.
- Right: Supplier Response Funnel: Requested → In Progress → Submitted → Accepted. Conversion rate per stage.

Row 4 — AI Insights:

- AI Predictive Compliance (AI-6): 12-month risk trajectory chart. Red/amber/green forecast.
- Regulatory Alert Feed (AI-4): Latest ECHA/REACH/CLP/GADSL changes with BOM impact count.

5.2 MDS Review

MDS Review Queue *Route: /oem/mds*

Table of all MDS submitted by suppliers, pending review.

- DataTable: Part #, Supplier, MDS Version, Submission Date, AI Risk Score, Status. Sort by risk score to prioritise.
- Tabs: All / Pending Review / Accepted / Rejected / Needs Update.
- AI Priority Ranking: Sort by AI-predicted risk score (AI-1) so highest-risk MDS reviewed first.

MDS Review Detail *Route: /oem/mds/[id]/review*

Full MDS review interface with approve/reject workflow.

- Split view: Left = MDS data (tree, substances, PCF), Right = Compliance check results.
- AI Prediction Panel: Compliance risk score with SHAP explanation of contributing factors.

- Rule-by-rule results: Expand each regulation to see pass/fail per substance.
- Comment thread: Add review comments visible to supplier.
- Actions: Accept, Reject (with reason), Request Update (specify which fields).
- Version diff: Compare against previous accepted MDS version.

5.3 BOM Analysis

BOM Analysis *Route: /oem/bom*

Upload and analyse Bills of Materials for compliance across 100k+ parts.

- BOM Upload: CSV, XML, or SAP sync. Support design-BOM, orderable, stock (RFI §7).
- Stored BOMs: Select from previously uploaded BOMs per business function.
- Analysis Results: Filterable table with RAG status per part, PFAS flags, SVHC %, PCF data.
- Performance: <100ms via CQRS read model (OpenSearch + Redis).
- Drill-down: Click part → substance-level view with regulation-specific flags.
- Historical Analysis: Re-analyse BOM against rules in force at any point in time (RFI §7).
- Export: PDF/Excel/CSV compliance report per BOM.

5.4 SDS Archive

SDS Management *Route: /oem/sds*

Archive and workflow for 17,000+ SDS PDFs.

- Searchable archive with OCR full-text search. Filter by substance, mixture, region, language.
- Cross-region access: SDS collected in any region viewable by all regions (RFI §2).
- Workflow queue: Draft → Under Review → Approved/Rejected with comments.
- Metadata: Where mixture is used (organisation, department, section), how used, multiple locations.
- SDS → MDS Link: View linked MDS data sets. Generate XML from SDS (RFI §2, §9).
- Translation: AI translation status, confidence scores, 26-language support.

5.5 SCIP Module

SCIP/ECHA *Route: /oem/scip*

SCIP dossier management and ECHA submission tracking.

- Dossier List: All SCIP dossiers with submission status (Draft/Submitted/Confirmed/Error).
- Dossier Builder: Auto-generate IUCLID i6z from MDS data. Edit concentration ranges, TARIC codes.
- TARIC Code Manager: Import/maintain customs codes per article (RFI §14).
- Bulk Submission: Submit multiple dossiers via ECHA system-to-system API.
- Simplified SCIP: For different legal entities of registered uses (RFI §14).
- UUID Tracking: ECHA-assigned UUIDs and candidate package references.

5.6 REACh Module

REACh *Route: /oem/reach*

Complete REACh registration, restriction, and authorisation management.

- Registration Tracker: All substances/mixtures by legal entity with tonnage bands.

- Volume Monitor: Alert when substance volumes cross 1t/10t thresholds. Import from XLS/XML (RFI §12).
- Entity Management: Track entities using substances, how used, import/manufacture status.
- Registered Uses Report: Confirm Nissan use aligns with registered uses.
- Restriction List: Annex XVII restricted substances with dates, restricted uses (RFI §13).
- Authorisation: Annex XIV substances with sunset dates, authorised uses.
- SVHC Tracker: Import/manufacture volumes per legal entity per SVHC. Auto-generate ECHA notification.
- Direct Material Volumes: Upload volumes per vehicle for total SVHC determination (RFI §6).
- Communication Obligation: Flag products with SVHC >0.1% w/w.
- Notification Obligation: Combine SVHC content + volumes to report by legal entity.

5.7 CLP Module

CLP Route: /oem/clp

Classification, Labelling and Packaging management.

- Hazard Classification: Identify hazardous substances (SDS Section 2) and non-hazardous mixtures containing hazardous substances (Section 3) (RFI §15.1-2).
- CLP Notification Candidates: Maintain candidate list (RFI §15.4).
- Label Generator: Auto-generate GHS labels with pictograms, signal words, H/P statements in 26 languages. Standard + small package variants (RFI §15.5, 15.7).
- Label Upload: Upload final labels of marketed products (RFI §15.5).
- Search/Display: Full search and download for all CLP information (RFI §15.6).
- PCN (Poison Centre Notifications): UFI generator, IUCLID i6z dossier builder, ECHA PCN portal connector (RFI §15.8).

5.8 RRR Module

RRR Route: /oem/rrr

Reusability, Recyclability, Recoverability per ISO 22628.

- Material Taxonomy: Classify materials as metallic/polymeric/fluid/electronics/other with recyclable/recoverable/disposable flags.
- RRR Calculator: Mass-based recyclability (≥85%) and recoverability (≥95%) per vehicle model.
- Recycled Content Tracker: Pre/post-consumer %, ELV-origin %. Target: 25% recycled plastic.
- Critical Raw Materials: Track CRMs, plastics, steel, aluminium recovery.
- Certificate Generator: RRR compliance certificates for type-approval.

5.9 Sustainability Module

Sustainability Hub Route: /oem/sustainability

PCF, DPP, CBAM, CSRD — all sustainability features.

Sub-pages:

- PCF Dashboard (/oem/sustainability/pcf): Carbon intensity per product line, supplier, region. Drill to MDS-level PCF. IMDS 15.x weekly digest.
- DPP Generator (/oem/sustainability/dpp): Create/manage Digital Product Passports. LLM narrative generation (AI-5). Schema compliance checker.

- CBAM Tracker (/oem/sustainability/cbam): CN/TARIC codes, embedded emissions, CBAM certificate calculator.
- CSRD Export (/oem/sustainability/csrd): ESRS-format data export. Link SVHC volumes to Scope 3 emissions.
- Catena-X (/oem/sustainability/catena-x): EDC connector status, sovereign PCF data exchange, Pathfinder.

5.10 Supplier Management

Supplier Management *Route: /oem/suppliers*

Manage 4,000+ suppliers and drive compliance data collection.

- Supplier List: Searchable table with compliance score, response rate, risk tier, last activity.
- Chasing Engine (/oem/suppliers/chasing): AI-driven outreach (AI-3). Risk-scored priority list. Batch messaging. Rate control. Communication history.
- Response Funnel: Visual funnel: Requested → In Progress → Submitted → Accepted with conversion rates.
- Supplier Profile (/oem/suppliers/[id]): Contact details, submission history, compliance score trend, risk assessment, linked MDS/SDS.
- Management Reports: Suppliers not responded, MDS needing review by expiry date (RFI §1.3).

5.11 Reports

Reports & Analytics *Route: /oem/reports*

Comprehensive reporting engine.

- Standard Reports: Missing MDS, missing SDS, SVHC volumes, SVHC by legal entity, RRR calculations, tonnage reports (RFI §7).
- Report Builder: Drag-drop column selector, filter builder, save custom report templates.
- Scheduled Reports: Configure auto-generation (daily/weekly/monthly) with email delivery.
- Export: PDF, Excel, CSV, XML. Compliance-ready formatting.

5.12 Integrations

Integration Hub *Route: /oem/integrations*

Manage all 10 system interfaces from RFI §6.

- Integration cards: Each of 10 interfaces with status (Connected/Disconnected/Error), last sync time, config.
- IMDS Sync: Bi-directional sync status, PCF sync, CAMDS support.
- SAP ERP: BOM sync, production volumes, stock lists.
- ECHA: SCIP + PCN submission status.
- Webhook Management: Configure custom webhooks, view delivery log, test endpoints.
- API Keys: Generate/revoke API keys per integration.

5.13 AI Insights

AI Intelligence Centre *Route: /oem/ai-insights*

Central hub for all 6 AI innovations.

- Compliance Predictions (AI-1): Risk-ranked MDS list with SHAP explanations. Accuracy metrics.
- Regulatory Alerts (AI-4): Real-time feed of regulation changes with BOM impact analysis.

- Predictive Analytics (AI-6): 12-month forecasting dashboard. Deadline clusters, resource planning.
- Query Assistant: Natural language compliance queries ('Which suppliers have SVHC >0.1% in Qashqai parts?'). Text-to-SQL.
- Model Performance: Accuracy dashboards, drift alerts, model version history.

5.14 Settings

OEM Settings *Route: /oem/settings*

Organisation-level configuration.

- Organisation Profile: Legal entity details, regions (4-region Nissan structure).
- User Management: Add/remove users, assign roles (Compliance Engineer, Manager, Admin).
- Role Permissions: RBAC matrix editor. ABAC attribute policies (region, legal entity, product line).
- API Configuration: API keys, rate limits, webhook endpoints.
- Notification Rules: Configure alert thresholds, digest schedules, escalation paths.

6. Admin Portal — Page-by-Page Specification

7 sections. For Marklytics platform administrators managing multi-tenant SaaS infrastructure.

6.1 Admin Dashboard

Platform Dashboard *Route: /admin/dashboard*

System health and business metrics overview.

- System Health: API latency (p50/p95/p99), error rates, queue depths, active connections.
- Tenant Stats: Active tenants, user counts, MDS volume trends.
- Revenue: MRR, subscription tier distribution, usage metering.
- Infrastructure: AWS cost breakdown, resource utilisation, FinOps metrics.

6.2 Tenant Management

Tenants *Route: /admin/tenants*

Multi-tenant provisioning and configuration.

- Tenant List: Name, tier (Starter/Professional/Enterprise), user count, DB type (shared/dedicated), status.
- Tenant Detail (/admin/tenants/[id]): Configuration, feature flags, resource limits, DB isolation toggle.
- Provisioning Wizard: Create new tenant with tier selection, DB provisioning, admin user setup.
- Enterprise DB Isolation: Promote tenant from shared to dedicated RDS instance.

6.3 User Management

Users *Route: /admin/users*

Cross-tenant user administration.

- Global user list: Name, email, tenant, role, last login, MFA status.
- SAML/SSO Configuration: Per-tenant SAML provider setup (Azure AD, Okta, ADFS).
- Role Templates: Pre-defined role templates per portal.

6.4 Billing

Billing *Route: /admin/billing*

Subscription management and usage metering.

- Subscription Overview: Tier distribution, upcoming renewals, churn risk.
- Invoice Management: Generate, send, track payment status.
- Usage Metering: Per-tenant usage (API calls, storage, AI inference minutes).

6.5 System

System Health *Route: /admin/system*

Infrastructure monitoring and management.

- Service Health: Per-service status (API, workers, AI/ML, connectors). Green/yellow/red.
- Queue Monitor: SQS/EventBridge queue depths, dead-letter queue items.
- Job Scheduler: Running/pending/failed background jobs. Retry failed jobs.
- Feature Flags: Toggle features per tenant (PostHog integration).

6.6 AI Model Management

AI Models *Route: /admin/ai-models*

MLflow model registry administration.

- Model Registry: All 6 AI models with versions, stage (Staging/Canary/Production), metrics.
- A/B Tests: Active experiments, traffic splits, performance comparison.
- Model Cards: View/edit model documentation, bias assessments, EU AI Act classification.
- Drift Monitoring: Evidently AI dashboards. Data drift + concept drift alerts.
- Retraining Pipeline: Trigger manual retraining, view pipeline status.

6.7 Audit Logs

Audit Trail *Route: /admin/audit-logs*

Immutable event log for SOC 2 compliance.

- Event Log: Searchable, filterable log of all system actions. User, action, resource, timestamp.
- Filters: By user, action type, resource type, date range, tenant.
- Export: CSV/JSON export for compliance auditors.
- Retention: Configurable retention period per tenant (default: 7 years).

7. Compliance Officer Portal — Page-by-Page Specification

5 sections. For senior compliance staff needing executive oversight, regulatory forecasting, and audit evidence.

7.1 Executive Dashboard

Compliance Dashboard *Route: /compliance/dashboard*

High-level compliance program status for senior leadership.

- Program Score: Single number 0-100 representing overall compliance health.
- Risk Heatmap: Aggregated across all product lines and regulations.
- Trend Lines: 12-month compliance score trend, supplier response rate trend.
- Top Risks: Top 10 non-compliant parts/substances requiring attention.
- Coverage Metrics: MDS coverage, SDS coverage, SCIP coverage, PCF coverage.

7.2 Regulatory Intelligence

Regulatory Tracker *Route: /compliance/regulatory*

Real-time regulatory change monitoring powered by AI-4.

- Alert Feed: Chronological feed of regulation changes (ECHA, REACH, CLP, GADSL, PFAS, ELV).
- Impact Analysis: Per alert, show affected BOMs, parts count, suppliers impacted.
- Severity Classification: Critical / High / Medium / Informational.
- Action Items: Auto-generated recommended actions per change.
- Subscription Manager: Configure which regulation feeds to monitor.

7.3 Forecasting

Predictive Compliance *Route: /compliance/forecasting*

AI-powered forward-looking compliance intelligence (AI-6).

- 12-Month Risk Trajectory: Interactive chart showing predicted compliance score over next year.
- Deadline Cluster Analysis: Identify months with concentrated regulatory deadlines.
- Resource Planning: AI-recommended staffing needs based on predicted workload.
- What-If Scenarios: Simulate impact of new regulation / SVHC addition / supplier loss.

7.4 Digital Product Passports

DPP Management *Route: /compliance/dpp*

EU Battery/Vehicle Regulation DPP lifecycle.

- DPP List: All issued passports with status, vehicle model, issuance date.
- DPP Generator: Select vehicle/product → AI generates narrative (AI-5) → human review → issue.
- Schema Validator: Check DPP against EU regulation schema before issuance.
- Audit Trail: Full history of DPP creation, edits, approvals.

7.5 Audit & Evidence

Compliance Audit *Route: /compliance/audit*

Audit-ready evidence management.

- Evidence Packages: Pre-assembled evidence bundles per regulation for auditors.
- Certificate Repository: RRR certificates, SCIP confirmations, REACh registrations.
- Compliance Timeline: Chronological view of all compliance decisions and submissions.
- Export: Full audit package export (PDF + attachments) for third-party auditors.

8. Complete Page Inventory (All Portals)

Total unique pages/routes across all 4 portals:

Portal	Route	Page Description
Auth	/login	Login (email + SSO)
Auth	/register	Supplier self-registration
Auth	/forgot-password	Password reset
Auth	/sso-callback	SAML SSO callback
Supplier	/supplier/dashboard	Supplier dashboard
Supplier	/supplier/mds	MDS list
Supplier	/supplier/mds/new	MDS creation wizard (6 steps)
Supplier	/supplier/mds/[id]	MDS editor
Supplier	/supplier/mds/[id]/tree-builder	Material tree builder
Supplier	/supplier/mds/[id]/validate	Validation & AI prediction
Supplier	/supplier/sds	SDS list & upload
Supplier	/supplier/sds/[id]	SDS detail & workflow
Supplier	/supplier/sui	SUI management
Supplier	/supplier/documents	Document library
Supplier	/supplier/requests	OEM requests inbox
Supplier	/supplier/profile	Supplier profile & settings
OEM	/oem/dashboard	OEM executive dashboard
OEM	/oem/mds	MDS review queue
OEM	/oem/mds/[id]/review	MDS review detail
OEM	/oem/bom	BOM management
OEM	/oem/bom/analysis	BOM analysis results
OEM	/oem/bom/upload	BOM upload
OEM	/oem/bom/[id]	BOM drill-down
OEM	/oem/sds	SDS archive & workflow
OEM	/oem/scip	SCIP dossier management
OEM	/oem/scip/builder	SCIP dossier builder
OEM	/oem/reach	REACH overview
OEM	/oem/reach/registrations	REACH registrations
OEM	/oem/reach/volumes	REACH volume monitor
OEM	/oem/reach/restrictions	REACH restrictions & authorisations
OEM	/oem/reach/svhc	SVHC tracker
OEM	/oem/clp	CLP overview
OEM	/oem/clp/labels	CLP label generator
OEM	/oem/clp/pcn	PCN dossier management
OEM	/oem/rrr	RRR calculator & taxonomy
OEM	/oem/rrr/certificates	RRR certificates
OEM	/oem/sustainability/pcf	PCF dashboard
OEM	/oem/sustainability/dpp	DPP generator

OEM	/oem/sustainability/cbam	CBAM tracker
OEM	/oem/sustainability/csrd	CSRD export
OEM	/oem/sustainability/catena-x	Catena-X EDC
OEM	/oem/suppliers	Supplier list
OEM	/oem/suppliers/chasing	AI chasing engine
OEM	/oem/suppliers/[id]	Supplier profile
OEM	/oem/reports	Reports & analytics
OEM	/oem/reports/builder	Custom report builder
OEM	/oem/integrations	Integration hub
OEM	/oem/ai-insights	AI intelligence centre
OEM	/oem/ai-insights/query	Query assistant
OEM	/oem/settings	OEM settings & users
Admin	/admin/dashboard	Platform dashboard
Admin	/admin/tenants	Tenant management
Admin	/admin/tenants/[id]	Tenant detail & config
Admin	/admin/users	User management
Admin	/admin/billing	Billing & subscriptions
Admin	/admin/system	System health & jobs
Admin	/admin/ai-models	AI model registry
Admin	/admin/audit-logs	Audit trail
Compliance	/compliance/dashboard	Executive dashboard
Compliance	/compliance/regulatory	Regulatory intelligence
Compliance	/compliance/forecasting	Predictive forecasting
Compliance	/compliance/dpp	DPP management
Compliance	/compliance/audit	Audit & evidence

Total: 63 unique page routes across 4 portals + authentication.

9. Frontend Implementation Roadmap

Aligned with the 24-month backend roadmap from the architecture document:

Phase 1 — Foundation & Shell (Months 1–4)

- Next.js 15 project scaffold with App Router and RSC.
- Design system setup: Tailwind 4, shadcn/ui, tokens, Storybook.
- AppShell: Sidebar, TopBar, Breadcrumbs, CommandPalette.
- Authentication: Login, SSO callback, forgot password, session management.
- Role-based routing: Portal selection based on user role.
- i18n: next-intl setup with English + 2 pilot languages.
- Core components: DataTable, FileUpload, StatusBadge, EmptyState, KpiCard.
- OEM Dashboard: KPI cards, placeholder heatmap.
- Deliverable: Working shell with auth, navigation, and OEM dashboard prototype.

Phase 2 — Supplier Portal (Months 5–8)

- Supplier Dashboard: All 5 panels live.
- MDS Creator: Full 6-step wizard with tree builder (dnd-kit), substance entry, AI classification.
- SDS Upload: Drag-drop with OCR preview.
- SUI Management: Upload and linking.
- Requests Inbox: Priority-sorted OEM requests.
- Document Library: Grid/list view, full-text search.
- Supplier Profile: Settings, language preference.
- Real-time: WebSocket notifications for chasing messages.
- Deliverable: Full supplier portal ready for pilot.

Phase 3 — OEM Portal Core (Months 9–12)

- OEM Dashboard: Full implementation with live heatmap, deadline timeline, funnel.
- MDS Review: Queue + detail view with AI prediction panel.
- BOM Analysis: Upload, analysis results, drill-down, historical.
- SDS Archive: Searchable archive with workflow queue.
- SCIP Module: Dossier builder, submission, TARIC manager.
- REACH Module: Registration tracker, volumes, restrictions, SVHC.
- CLP Module: Hazard classification, label generator, PCN builder.
- RRR Module: Calculator, taxonomy, certificates.
- Supplier Management: List, chasing engine, response funnel.
- Reports: Standard reports, export engine.
- Deliverable: Full OEM portal with all RFI modules.

Phase 4 — AI & Sustainability (Months 13–18)

- AI Insights Centre: Predictions, alerts, analytics, query assistant.
- Sustainability Module: PCF dashboard, DPP generator, CBAM, CSRD, Catena-X.
- Integration Hub: All 10 interfaces with live status and configuration.
- Report Builder: Drag-drop custom reports, scheduled delivery.

- Compliance Officer Portal: All 5 sections live.
- AI visualisations: SHAP explanations, risk trajectories, regulatory impact charts.
- Deliverable: AI-powered features and sustainability modules live.

Phase 5 — Admin & Polish (Months 19–24)

- Admin Portal: All 7 sections (tenants, users, billing, system, AI models, audit).
- i18n completion: All 26 languages.
- Accessibility audit: WCAG 2.1 AA compliance across all portals.
- Performance: Lighthouse 90+ scores, Core Web Vitals optimisation.
- E2E Tests: Playwright tests for all critical user flows.
- Mobile optimisation: Responsive refinement for all portals.
- Deliverable: Production-ready, enterprise-grade frontend.